Sources of error

The colourimetry and survey data collected for this repository has sources of error due to lack of information, human errors, and biases in the interviews.

During the production of the samples and measuring of the three sections the students had no prior knowledge of the assignment. Due to the lack of information of the intended use for the sample, some of the students painted the samples unevenly, which resulted in visible pain strokes and patchiness of the colour. This affected the survey as the first question aimed to observe whether the participants would notice the three areas without prior knowledge of the experiment, as the patchiness created confusion and several participants could not separate the patchiness with the three areas. Several of the students mentioned having to clarify the three areas through the rest of the questions, which made the questioning fairly biased. The questioning would also be less biased if executed in the same lighting for each interview with stricter guidelines for the questioning.

Five colourimetry measurements were taken. Since the use of the colourimetry data was unknown to the students, their placement of the spectrophometer did not cover the entire area making them random and imprecise. Caused by the uneven paint layers the measurements of the covered, uncovered and TruVue areas could result in inaccuracies. To improve the representation the number of measurements could have been increased, and the placement could be distributed more evenly over the entire sections.
All students except one painted with the same paint and brush, making one of the swabs different from the rest. The colourimetry data differs from the rest but has been included in the complete dataset and analysis. Additionally, it was noted that all the students had individual issues, making the data inadequate.

Some improvements could be done to the survey, as the questions and strategy of the interviews should have been arranged to reduce bias and make the data more comparable. The three areas were lined up, which made the first and the third section not as easily comparable as the others. This could affect the answers as the colour difference could be less noticeable. Instead of having them lined up, the students could have had three different swabs or cut them in three parts, so the participants could juxtapose them. Student 10 cut their sample into the three marked areas, but they stated none of the participants felt they needed to change their positions. However, this was because they saw no difference between the colours of this swab, which could differ from the rest of the swabs.